

Riphah International Colleges

A Project of Riphah International University

Course Outline

Course Code	CS2713
Course Title	Game Programming
Credit Hours	3(2+1)
Prerequisites	N/a
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	Quizzes + Assignments: 10 Project+ Presentation 10 Lab : 15 Mid Term 25 Final Exam: 40
Course Description	The ultimate goal of game programming is to deliver an entertaining, visually appealing, and well-performing game that captivates players and provides an enjoyable and memorable experience. This involves a combination of technical skills, creativity, and collaboration within a game development team
Textbook	"Game Programming Patterns" – Robert Nystrom "Introduction to Game Design, Prototyping, and Development" – Jeremy Gibson Bond
Reference Material	Lecture Slides, notes, "The Art of Game Design: A Book of Lenses" – Jesse Schell
Course Goals/Objectives	Goal: • Creating Engaging Gameplay • Implementing Game Mechanics • Optimizing Performance • Graphics and Visual Effects • Audio Integration • Network Programming (for Online Multiplayer Games)

Topics Covered in the Course, with Number of Lectures on Each Topic

Topics	Hours
Introduction to Game programming	3
Game engine	12
Game Level Design,	14
Game world	6
Game Technology	3

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Game Documentation	4
Business for gaming	6

Weekly Lecture Plan

Week	Lecture	Topic
1	1	Introduction to Game programming
		Game Design and Development 1: 2D Shooter
	2	The game process How are game made
		Game designer role The game Development team Costs and limitation
2	3	Game engine overview Solar system project overview Download and install Unity platform
		Unity setup Create unity project Download project assists
	4	import Assists Unity editor
3	5	Game engine overview Solar system project overview Download and install Unity platform
		Unity setup Create unity project Download project assists Materials, Lights, Behaviors and Cameras
	6	Audio, Prefabs, Build and test Finishing up Solar system simulation Setting up the project Level Design Game system
4	7	Level Design Game system
		taking screenshot Additional Resources
	8	Create new unity project
5	9	Get new input system Player setup visual
		Player setup control Player setup health damage
	10	Player setup fighting backup
6	11	Cameras Framing the Action Level Setup Environment Sprites

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		Game Loop Setup 1 Management and UI Game Loop setup 3 In Game UI and main Menu
	12	Audio and Music Adding More levels Code Basics
	13	Unity Programming_ Introduction Unity Programming_ Object Oriented Thinking Unity Programming_ Anatomy of C# Script
7	14	C# Variables Conditionals and Loops Functions and Classes C# within Unity Programming in Unit
	15	Tilemaps_ Tile Palette and Tiles Tilemaps_ Rules Tiles Animated Tiles
8	16	Combining Rules and Animation Player setup Physics Player setup input System Canvases and user interface Level Setup
MID TERM EXAM		
	17	Introduction Game word Components Designing the game world
9	18	World Building Story telling Introduction character and Documentary Character Design Setting and Plot
	19	Game play introduction Decomposing Gameplay The secret of Mario's jump(and other versatile verbs)
10	20	forms of fun Game Genres user experience inputs outputs The mechanics of Movement Designing with UX
	21	Game technology Introduction Brief History Game technology Hardware Game technology Software
11	22	Game technology Environment A brief history of video game
12	23	3D Shooter Game Project setup Project Overview

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		Creating A New Unity Project Getting The New Input System
		Main Menu Setup 1 - Creating Menu Main Menu Setup 2 - About page Main Menu Setup 3 - Dynamic elements Main Menu Setup 4 - Unity terrain Creating First Level Player setup
	24	Writing The Player Controller Script 1, 2, 3 Enemy AI Setup 1 Boss Fight Game Graphics Game Graphics Concepts - Part 1 of 3 Game Audio Game Audio Concepts
		Programming Best Practices - Part 1: Introduction Programming Best Practices - Part 2: Easy-to-Read Game Balancing
	25	Game Balancing player vs player Player vs Game Play Game Play vs Game play
13	26	Game designing challenges Prototyping and Playtesting
		Game Play prototyping
	27	Other early prototyping Mid to late Prototypes
14	28	Playtesting
		Level Whiteboxing with ProBuilder 1 Level Whiteboxing with ProBuilder 2 Level Whiteboxing with ProBuilder 3
	29	The Art Pass 1 The Art Pass 2 Adding The Player And Animations Using the Provided Prefabs 1
15	30	Creating a Special Effect Ideation - Part 1: Introduction Changing the Way You Look at the World Ideation - Part 2: Brainstorming Warm Up Ideation - Part 3: Structured Brainstorming Design Documentation
		Business for Gaming Game Studio Budget Calculator
	31	Business for Gaming - Part 2: How are games funded? Business for Gaming - Part 3: Where are games sold? Business for Gaming - Part 4: How is money made on games?
16	32	Social Issues Building your Game Design Library What Game Development Role Interests You Most?

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17	33	Project/Presentation
	34	Project/Presentation
FINAL TERM EXAM		